

High-frequency data and limit order books



Lecture 7 - March 24th, 2026

Ioane Muni Toke

MICS Laboratory and Chair of Quantitative Finance
CentraleSupélec – Université Paris-Saclay, France



CentraleSupélec

U. Paris-Saclay CentraleSupélec cursus Ingénieur 3A Mathématiques et Data Science
IP Paris ENSAE M2 Statistics, Finance and Actuarial Science
U. Paris-Saclay Evry M2 Quantative Finance

Table of contents

Cryptocurrencies

Automated market makers

MEV attacks and Uniswap microstructure

Lab 6

Table of contents

Cryptocurrencies

Automated market makers

MEV attacks and Uniswap microstructure

Lab 6

Cryptocurrencies

- ▶ A **cryptocurrency** is a digital asset designed to be traded electronically on a distributed computer network.
- ▶ Ownership of the asset and transactions are stored electronically in a **blockchain**, a distributed, append-only ledger replicated across many nodes.
- ▶ Security relies on **cryptography**. Transactions can occur worldwide, 24/7, anonymously/pseudonymously, without banking/institutional intermediaries and are publicly displayed on the blockchain.
- ▶ Bitcoin (BTC) and Ether (ETH) are the most well-known **coins** and have the largest market capitalization.
- ▶ **Tokens** are digital assets built on an existing blockchain. **Stablecoins** are tokens pegged to a fiat currency (USD) via a collateral (or algorithmic) mechanism. USDC, USDT, USDS are well-known stablecoins.

Centralized exchanges

- ▶ A **centralized exchange (CEX)** is a trading platform operated by a company on which clients can trade cryptocurrencies and digital assets (e.g. Binance, Coinbase, Kraken).
- ▶ Centralized exchanges typically implement **limit order books** to store and match client orders. Trading rules similar to the standard LOB architecture on equity.
- ▶ CEXs are the first type of crypto exchanges by trading volume. A CEX is a **bank/broker/exchange/custodian** hybrid but **without regulations and without insurances**.
- ▶ **Risk of market manipulation** (no administrative/institutional supervision): wash trading, pump-and-dump, ...
- ▶ **Custodial risk**: client funds are held by unregulated and uninsured private companies.
Famous CEX bankruptcies: Mt.Gox (2014, fraud, criminal conviction), FTX (2022, fraud, criminal conviction), Celsius Network (2022, fraud, criminal conviction), ...

Decentralized exchanges

- ▶ A **decentralized exchange** (DEX) is a trading protocol running entirely on a blockchain via **smart contracts**, programs that are executed on a blockchain.
- ▶ Users trade directly from their own wallets: transaction settlement is on-chain.
- ▶ The dominant DEX architecture is not based on a limit order book, but rather on a **liquidity pool** and an automated market maker (AMM), a pricing formula that governs the prices of transactions (see the example of Uniswap in the next section).
- ▶ Every operation on-chain incurs a **gas fee** paid to the network. Fees depend on network congestion. On top of a base fee, user can choose to add a priority fee.
- ▶ Uniswap (Ethereum) and PancakeSwap (BNB Chain) are well-known decentralized exchanges. The **total value locked** (TVL) is the amount of assets deposited in a DeFi protocol.

Table of contents

Cryptocurrencies

Automated market makers

MEV attacks and Uniswap microstructure

Lab 6

Automated market makers

- ▶ An **automated market maker** (AMM) replaces the limit order book with a **liquidity pool** and a deterministic **pricing rule**.
- ▶ The liquidity pool is a reserve of two tokens X and Y . Market participants can deposit tokens and earn fees for providing liquidity (**liquidity providers**). They can also trade (swap) one token for the other (**liquidity takers**).
- ▶ The fee charged on every swap and distributed to liquidity providers is usually a fixed fraction φ of the traded volume (0.01%, 0.05%, 0.3%, ...).
- ▶ Amounts deposited or swapped satisfy a pricing rule. In Uniswap, a **constant product market maker** (CPMM) is used: $q^X q^Y = \text{const}$.
- ▶ Other AMMs include constant sum market maker ($q^X + q^Y = \text{const}$), hybrid formulas ($A(q^X + q^Y) + q^X q^Y = \text{const}$), CPMM on weighted pools ($\prod_i q_i^{w_i} = \text{const}$),...

Uniswap v2

- ▶ Protocol launched in 2020 [1].
- ▶ Each pool is identified by two tokens X and Y and a fee level φ .
- ▶ q_t^X and q_t^Y are the **reserves** of tokens X and Y at time t .
- ▶ The **total liquidity** of the pool is $L_t = \sqrt{q_t^X q_t^Y}$.
- ▶ Market participants interact with the pool with **three types of events**:
 - ▶ **Swap**: an agent (liquidity taker) deposits x units of X and receives y units of Y (or vice versa) ;
 - ▶ **Mint**: an agent (liquidity provider) adds liquidity by depositing both tokens, x units of X and y units of Y ;
 - ▶ **Burn**: an agent (liquidity provider) removes previously deposited liquidity of both tokens.

Uniswap v2 – Swap I

- Swap: an agent deposits x units of X in order to receives y units of Y .
- The CPMM rule states that the product of reserves must be preserved (net of fees) after every swap:

$$(q_t^X + (1 - \varphi)x)(q_t^Y - y) = q_t^X q_t^Y = L_t^2,$$

so that the agent receives

$$y = x \frac{(1 - \varphi) q_t^Y}{q_t^X + (1 - \varphi)x}.$$

- The **average transaction price** of X in terms of Y (numeraire) is

$$\frac{y}{x} = \frac{(1 - \varphi) q_t^Y}{q_t^X + (1 - \varphi)x} \xrightarrow{\varphi \rightarrow 0} \frac{q_t^Y}{q_t^X + x}.$$

Uniswap v2 – Swap II

- ▶ The **marginal price** of X in terms of Y is obtained with $x \rightarrow 0$, i.e. is

$$p_t = \frac{q_t^Y}{q_t^X},$$

and is called **pool price** or reference price.

- ▶ In Uniswap v2, fees are not cashed out to be immediately distributed, but are compounded, i.e. included in the pool to increase liquidity: after the swap the actual reserves are updated to

$$q_{t+}^X = q_t^X + x, \quad q_{t+}^Y = q_t^Y - y$$

i.e. $L_{t+}^2 > L_t^2$.

- ▶ The **post-swap pool price** is $p_{t+} = \frac{q_{t+}^Y}{q_{t+}^X} = \frac{q_t^Y - y}{q_t^X + x} < \frac{q_t^Y}{q_t^X} = p_t$.

Uniswap v2 – Mint and burn

- ▶ **Mint:** an agent deposits both x and y to increase the pool liquidity.
- ▶ The CPMM rule states that liquidity must be added in exact proportion to current reserves so that the pool price does not change:

$$p_{t+} = \frac{q_t^Y + y}{q_t^X + x} = \frac{q_t^Y}{q_t^X} = p_t \quad \text{i.e.} \quad \frac{y}{x} = \frac{q_t^Y}{q_t^X} = p_t$$

- ▶ After the mint, the pool liquidity satisfies

$$L_{t+}^2 = (q_t^X + x)(q_t^Y + y) = (L_t + \sqrt{xy})^2,$$

i.e. $\sqrt{xy}$ is the added liquidity credited to the mint.

- ▶ **Burn:** an agent removes both x and y to redeem (part of) their share of the pool. Formulas identical to the mint operation, but with negative x and y .

Uniswap v3 – Concentrated liquidity

- ▶ Protocol launched in 2021 [2].
- ▶ In Uniswap v3, liquidity providers (LP) can specify a price range in which the added liquidity is valid.
- ▶ Price ranges must be defined using a **discrete price grid** $\{1.0001^{j\Delta} : j \in \mathbb{Z}\}$ where Δ is the pool's `tickSpacing` parameter (e.g. $\Delta = 10$ for the 0.05% fee pool).
- ▶ Total available liquidity as a function of the price is called the **liquidity profile** of the pool. This liquidity profile may exhibit large variations across prices/ticks.
- ▶ **Active liquidity** is the total liquidity available at the current price p_t .
- ▶ Fees in v3 are **non-compounding**: they are stored in a separate fee accumulator and not reinvested into the pool.

Uniswap v3 – Swap

- ▶ Swap: an agent deposits x units of token X in order to receive y units of token Y .
- ▶ Current pool price is p_t , active liquidity is L_t in range $[p_l, p_u]$. As long as the price does not reach the boundaries, Uniswap v2 CPMM rules apply, but without fee compounding:
 - ▶ quantity received: $y = x \frac{(1-\varphi) q_t^Y}{q_t^X + (1-\varphi)x}$;
 - ▶ post-swap price: $p_{t+} = \frac{q_{t+}^Y}{q_{t+}^X} = \frac{q_t^Y - y}{q_t^X + (1-\varphi)x} < \frac{q_t^Y}{q_t^X} = p_t$
- ▶ The lower boundary p_l is reached for the amount $\bar{x}$ that solves $p_{t+} = p_l$:

$$\bar{x} = \frac{L_t}{(1-\varphi)} \left(\frac{1}{\sqrt{p_l}} - \frac{1}{\sqrt{p_t}} \right).$$

- ▶ If $x \leq \bar{x}$, the swap is executed fully within $[p_l, p_u]$. If $x \geq \bar{x}$, $\bar{x}$ is executed first, then the remaining quantity $x - \bar{x}$ is executed as a new swap at the **next liquidity level**.

Uniswap v3 – Mint and burn

- ▶ Mint: an agent deposits both x units of token X and y units of token Y to increase liquidity in price range $[p_l, p_u]$.
- ▶ The CPMM formula with price range $[p_l, p_u]$ is

$$\left(x + \frac{\lambda}{\sqrt{p_u}}\right) (y + \lambda\sqrt{p_l}) = \lambda^2,$$

where λ is the liquidity credited to the mint.

- ▶ $\lambda = \sqrt{xy}$ if $p_l \rightarrow 0$ and $p_u \rightarrow +\infty$ (Uniswap v2 case).
- ▶ The mint algorithm first compute λ (straightforward) then defines mint quantities so that the current price is unchanged with

$$(x, y) = \begin{cases} \left(\lambda\left(\frac{1}{\sqrt{p_l}} - \frac{1}{\sqrt{p_u}}\right), 0\right) & \text{if } p_t < p_l, \\ \left(\lambda\left(\frac{1}{\sqrt{p_t}} - \frac{1}{\sqrt{p_u}}\right), \lambda(\sqrt{p_t} - \sqrt{p_l})\right) & \text{if } p_l \leq p_t \leq p_u, \\ (0, \lambda(\sqrt{p_u} - \sqrt{p_l})) & \text{if } p_t > p_u. \end{cases}$$

Table of contents

Cryptocurrencies

Automated market makers

MEV attacks and Uniswap microstructure

Lab 6

The Uniswap/Ethereum Transaction Pipeline

- ▶ The pipeline from the user to the blockchain is complex. [6, 4]:
Users → Mempool → Searchers → Builders → Relays → Validators → Blockchain.
- ▶ Transactions submitted by users first enter the public **mempool** (memory pool), a waiting room for unconfirmed transactions visible to all nodes.
- ▶ A **validator** is selected to propose the next block (one block every 12 seconds). The block is built by specialized **block builders** competing in a sealed-bid auction.
- ▶ On top of the blockchain standard fee/reward structure, profit can be made by **choosing and ordering transactions** in a block (**Maximal Extractable Value** (MEV) [3, 5]).
- ▶ MEV searchers and block builders scan the mempool to identify opportunities: arbitrages, JIT attacks, sandwich attacks, ...

Just-in-Time (JIT) Liquidity

- ▶ **Idea**: exploit a large swap by providing liquidity in a narrow range around the current price just before the swap executes in order to **capture (most of) the fee**, then close the position.
- ▶ **Structure** of a JIT attack:
 - ▶ Mint (attacker, narrow range, large L)
 - ▶ Swap (victim's transaction)
 - ▶ Burn (attacker, same range, same L)
- ▶ JIT may be seen as beneficial for the LT (lower price impact due to extra liquidity) but **hurts passive LPs** (fee revenue stolen).
- ▶ **Proposed detection**: in the same block, a (mint, burn) pair by the same wallet with identical range and liquidity amount, with at least one swap in between.
- ▶ **In the literature**: rare event, beneficial for user ([7], Uniswap foundation...) ; very frequent, 20% to 40% of minted liquidity in liquid pools, largely due to a very small number of wallets [4] ; ...

Sandwich attacks

- ▶ **Idea**: exploit the price impact of a large swap by **front-running** and then closing the position.
- ▶ **Structure** of a sandwich attack:
 - ▶ Front-run swap (attacker, same direction as victim, moves price unfavorably for the victim)
 - ▶ Victim swap (executes at a worse price than expected)
 - ▶ Back-run swap (attacker, opposite direction, closes position with a profit)
- ▶ Sandwich attacks are (extremely) **harmful to the victim**: they are forced to trade at a worse price.
- ▶ **Proposed detection**: in the same block, a (front-run, back-run) pair by the same wallet with at least one swap in between.
- ▶ **In the literature**: victim volume is found to be around 5% of all swapped volume in Uniswap v3[4].

Discussion

- ▶ DeFi promises (easy borderless access? transparency? no intermediaries?) vs. DeFi reality.
- ▶ Cryptocurrency trading on both CEXs and DEXs is subject to **price manipulation**.
- ▶ **Custodial risk** on CEXs is higher in the absence of regulations and insurances (fraud).
- ▶ **Multiple intermediaries** between a user and the blockchain. **Multiple fee structures** with bribes (priority in DEXs can be bought/bid for).
- ▶ MEV strategies could be seen as an **economic rent** benefiting **a very small number of actors**.
- ▶ The most profitable crypto companies are arguably stablecoin issuers (Tether) and CEXs (Binance, Coinbase), with very traditional financial models (**earning interests, or earning fees**).
- ▶ ...

Table of contents

Cryptocurrencies

Automated market makers

MEV attacks and Uniswap microstructure

Lab 6

Lab 6 - Questions

The new dataset must be place in the `./Data/` directory, with other `csv.gz` files. You may propose other public datasets relevant to the lecture (in that case, you must make the available to me).

1. Explore the proposed dataset. Comment, explain.
2. Detect MEV attacks such as JIT liquidity and sandwich attacks. Relate your results to the ones proposed in [4]. Comment, explain.

References I

- [1] H. Adams, N. Zinsmeister, and D. Robinson. *Uniswap v2 Core*. Tech. rep. Uniswap, 2020.
- [2] H. Adams, N. Zinsmeister, M. Salem, R. Keefer, and D. Robinson. *Uniswap v3 Core*. Tech. rep. Uniswap, 2021.
- [3] P. Daian, S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels. “Flash boys 2.0: Frontrunning, transaction reordering, and consensus instability in decentralized exchanges”. In: *arXiv preprint arXiv:1904.05234* (2019).
- [4] D. M. Di Nosse, F. Gatta, F. Lillo, and S. Jaimungal. “Deviations from Tradition: Stylized Facts in the Era of DeFi”. In: *arXiv preprint arXiv:2510.22834* (2025).
- [5] H. Materwala, S. M. Naik, A. Taha, T. A. Abed, and D. Svetinovic. “Maximal extractable value in decentralized finance: Taxonomy, detection, and mitigation”. In: *IEEE Transactions on Services Computing* (2025).

References II

- [6] B. Öz, D. Sui, T. Thiery, and F. Matthes. “Who wins ethereum block building auctions and why?” In: *arXiv preprint arXiv:2407.13931* (2024).
- [7] X. Wan and A. Adams. “Just-in-time liquidity on the uniswap protocol”. In: *Available at SSRN 4382303* (2022).